



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
Student Name	N. Siri Chandana Reddy
Roll Number	2520030142
Branch	CSE
Course Outcome	CO-1 (AVL TREE)

Case Study Topic: Waste Sense – Smart Waste Collection & Recycling Optimization System Using AVL

Work Done Summary:

The **Waste Sense – Smart Waste Collection & Recycling Optimization System** uses an **AVL Tree** to manage smart waste-bin records efficiently.

- Developed an AVL Tree-based data management system for storing waste-bin information.
- Inserted smart-bin IDs into the AVL Tree while maintaining balance through AVL rotations.
- Enabled fast searching of waste-bin records for monitoring and collection purposes.
- Implemented efficient insertion and deletion operations to handle newly added or removed bins.

Implementation Details:

1. **AVL Tree Creation:** An AVL Tree is created to store and manage smart waste-bin records using unique Bin IDs.
2. **Bin Record Insertion:** New waste-bin IDs are inserted into the AVL Tree following Binary Search Tree (BST) rules.
3. **Balance Maintenance:** After each insertion or deletion, the balance factor is checked and AVL rotations (LL, RR, LR, RL) are performed when necessary.
4. **Fast Searching:** The AVL Tree enables quick searching of waste-bin information with a time complexity of $O(\log n)$.
5. **Record Deletion and Updates:** Waste-bin records can be updated or deleted efficiently while maintaining the balanced structure of the tree.
6. **Data Display:** In order traversal is used to display all waste-bin records in sorted order for easy monitoring and management.

Result: Screenshot/output:

```
PS C:\Users\N.SiriChandana\OneDrive\Documents\DSA 23> java WasteSense
BFS Traversal:
Depot Bin1 Bin2 Bin3 RecycleCenter

Shortest Routes from Depot:
RecycleCenter : 6
Bin3 : 3
Bin2 : 2
Bin1 : 4
Depot : 0
```

The Waste Sense system successfully stores and manages smart waste-bin records using an AVL Tree. Bin IDs are inserted, searched, updated, and displayed efficiently in sorted order. The tree remains balanced after every operation, ensuring fast access to data.

Time Complexity

- Search Operation: $O(\log n)$
- Insertion Operation: $O(\log n)$
- Deletion Operation: $O(\log n)$
- In order Traversal: $O(n)$

GIT HUB LINK : <https://github.com/nemaliscreddy78-glitch/DSA2>

Student Signature

Faculty Signature